

Lab Work: Dictionary in Python

1. Online Shopping Order Analytics

Problem Statement

An e-commerce company stores product sales data as:

```
sales = {  
    "Laptop": 15,  
    "Mouse": 45,  
    "Keyboard": 32,  
    "Monitor": 12,  
    "Headphones": 28,  
    "Printer": 8,  
    "Webcam": 20,  
    "Speaker": 18,  
    "Tablet": 10,  
    "Router": 25  
}
```

Tasks

1. Display products sold more than 20 times.
2. Find the best-selling product.
3. Find the least-selling product.
4. Calculate total products sold.
5. Create a list of products requiring promotion (sales < 15).
6. Count products having sales between 10 and 30.

Sample Output

Products Sold More Than 20 Times:

Mouse
Keyboard
Headphones
Router

Best Selling Product: Mouse (45)

Least Selling Product: Printer (8)

Total Units Sold: 213

Products Requiring Promotion:

['Monitor', 'Printer', 'Tablet']

Products Having Sales Between 10 and 30: 6

2. Employee Performance Dashboard

Problem Statement

Employee performance scores are stored as:

```
performance = {  
  "EMP101": 92,  
  "EMP102": 78,  
  "EMP103": 45,  
  "EMP104": 88,  
  "EMP105": 97,  
  "EMP106": 56,  
  "EMP107": 81,  
  "EMP108": 64,  
  "EMP109": 39,  
  "EMP110": 73  
}
```

Tasks

1. Display employees scoring above 80.
2. Count employees needing improvement (score < 60).
3. Find the top performer.
4. Calculate average performance score.
5. Create separate lists:
 - Excellent (≥ 90)
 - Good (75–89)
 - Average (60–74)
 - Poor (< 60)

Sample Output

Employees Scoring Above 80:

EMP101
EMP104
EMP105
EMP107

Top Performer: EMP105 (97)

Employees Needing Improvement: 3

Average Score: 71.3

Excellent:

['EMP101', 'EMP105']

Good:

['EMP102', 'EMP104', 'EMP107']

Average:

['EMP108', 'EMP110']

Poor:
['EMP103', 'EMP106', 'EMP109']

3. City Temperature Monitoring System

Problem Statement

Daily temperatures of different cities are stored as:

```
temperature = {  
    "Delhi": 41,  
    "Mumbai": 33,  
    "Chennai": 37,  
    "Kolkata": 39,  
    "Bengaluru": 28,  
    "Pune": 30,  
    "Jaipur": 42,  
    "Lucknow": 40,  
    "Hyderabad": 35,  
    "Ahmedabad": 43  
}
```

Tasks

1. Display cities having temperature above 40°C.
2. Find the hottest city.
3. Find the coolest city.
4. Calculate average temperature.
5. Create a list of pleasant cities (temperature < 35°C).
6. Count cities with temperature between 35°C and 40°C.

Sample Output

Cities Above 40°C:

Delhi

Jaipur

Ahmedabad

Hottest City: Ahmedabad (43°C)

Coolest City: Bengaluru (28°C)

Average Temperature: 36.8°C

Pleasant Cities:

['Mumbai', 'Bengaluru', 'Pune']

Cities Between 35°C and 40°C: 4

4. Cricket Tournament Statistics

Problem Statement

Runs scored by players in a tournament:

```
runs = {  
    "Virat": 645,  
    "Rohit": 512,  
    "Gill": 698,  
    "Rahul": 435,  
    "Hardik": 278,  
    "Pant": 534,  
    "Surya": 389,  
    "Jadeja": 301,  
    "Iyer": 455,  
    "KL": 410  
}
```

Tasks

1. Display players scoring more than 500 runs.
2. Find the Orange Cap winner.
3. Find the lowest scorer.
4. Calculate total runs scored.
5. Create a list of players scoring below 400.
6. Count players scoring between 400 and 600 runs.

Sample Output

Players Scoring More Than 500 Runs:

Virat
Rohit
Gill
Pant

Orange Cap Winner: Gill (698)

Lowest Scorer: Hardik (278)

Total Tournament Runs: 4657

Players Scoring Below 400:

['Hardik', 'Surya', 'Jadeja']

Players Between 400 and 600 Runs: 5

5. Smart Electricity Billing System

Problem Statement

Monthly electricity consumption (units) is stored as:

```
units = {  
    "House101": 320,
```

```
"House102": 180,  
"House103": 510,  
"House104": 275,  
"House105": 150,  
"House106": 430,  
"House107": 220,  
"House108": 390,  
"House109": 145,  
"House110": 600  
}
```

Tasks

1. Display houses consuming more than 400 units.
2. Find the highest-consuming house.
3. Find the lowest-consuming house.
4. Calculate total units consumed.
5. Create lists:
 - Low Consumption (< 200)
 - Medium Consumption (200–400)
 - High Consumption (> 400)
6. Count houses eligible for an energy-saving campaign (consumption > 300).

Sample Output

Houses Consuming More Than 400 Units:

House103
House106
House110

Highest Consumption:

House110 (600 units)

Lowest Consumption:

House109 (145 units)

Total Units Consumed: 3220

Low Consumption:

['House102', 'House105', 'House109']

Medium Consumption:

['House101', 'House104', 'House107', 'House108']

High Consumption:

['House103', 'House106', 'House110']

Eligible for Energy-Saving Campaign: 5